

PDF and File Formats — Detailed Lecture Notes

JKSSB Computer Awareness

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33.1 What a file format is

Every file on a computer is stored as a long series of bits. On its own, a string of bits says nothing about whether it is a letter, a photograph, a spreadsheet, or a slide show. The **file format** is the rule that tells the software how those bits are arranged — where the text begins, where the image data sits, and how the page should be drawn. When a program opens a file, it reads the format to know how to interpret the contents.

The **extension** is the short suffix at the end of a file name, such as `.pdf`, `.docx`, or `.png`. The extension is a label that helps the operating system choose which program should open the file. It is convenient but only a name: the real identity of the file is its format.

Definition (File format). *The defined structure or rule by which data is stored in a file, so that software can correctly read, display, and save its contents.*

Definition (Extension). *The suffix at the end of a file name, usually written with a dot, that indicates the file's intended type (for example `.pdf`, `.docx`, `.png`).*

Term	Meaning in this lesson
Format	The actual structure of the data inside the file.
Extension	The dot-suffix in the file name, such as <code>.docx</code> .
Fixed layout	A format that stores exact positions, so the page looks the same everywhere.
Plain text	A file of ordinary characters with no formatting.

33.2 Extension versus format (the core trap)

The single most repeated confusion in this topic is treating the extension and the format as the same thing. They are not. In the file name `report.docx`, the base name is `report`, the extension is `docx` (written `.docx`), and the format is DOCX, the Office Open XML word-processing format. Likewise, `.png` is an extension, while PNG (Portable Network Graphics) is the format.

The extension can also be changed easily, and doing so does not convert the file. Renaming a photograph from `pic.jpg` to `pic.png` does not turn a JPEG image into a PNG image; it only changes the label. This is why the exam stresses the distinction: the extension is a name, the format is the content.

Example 33.2.1. *An item states, “`.png` is an image file format.” The statement mixes the two ideas. `.png` is the **extension**; PNG is the **format**. A correct statement would be, “PNG is an image file format, and `.png` is its usual extension.”*

33.3 Common document formats

The Office applications each save their work in a standard format. These are the Office Open XML formats, and each has its own extension.

Extension	Format	Typical use
.docx	Word document	Text, formatting, and images (MS Word)
.xlsx	Excel workbook	Worksheets, formulas, and charts (MS Excel)
.pptx	PowerPoint presentation	Slides and slide shows (PowerPoint)
.csv	Comma-Separated Values	Plain-text rows with values separated by commas
.txt	Plain text	Unformatted characters only

A **word-processing document** (DOCX) stores formatted text. A **workbook** (XLSX) stores worksheets with formulas and numbers. A **presentation** (PPTX) stores slides. A **CSV** file is plain text in which each line is a record and commas separate the fields; it is often used to move data between programs. A **TEXT** file holds plain characters and no formatting at all.

33.4 Common image extensions

Images also come in many formats, each suited to a different purpose. **PNG** (Portable Network Graphics) uses lossless compression and supports transparency, so it is good for logos and screen-shots. **JPG/JPEG** uses lossy compression, so it is good for photographs where a small loss of quality is acceptable. **GIF** (Graphics Interchange Format) handles simple images and short animations. **BMP** (Bitmap) stores a raw image and is usually larger. **TIFF** (Tagged Image File Format) is used for high-quality scanning and printing. **SVG** (Scalable Vector Graphics) stores a drawing as lines and shapes, so it scales without blurring.

Extension	Format	Character
.png	Portable Network Graphics	Lossless, transparency
.jpg	JPEG image	Lossy, photos
.gif	Graphics Interchange Format	Simple images, animation
.bmp	Bitmap	Raw, larger
.tiff	Tagged Image File Format	High quality, scanning/print
.svg	Scalable Vector Graphics	Vector, scales without blur

33.5 PDF: the Portable Document Format

PDF stands for **Portable Document Format**. It is a **fixed-layout** format: every item on the page is placed at a fixed position. Because the file also carries the fonts and images it needs, a PDF looks the **same on every device and printer**. This is why PDF is used for e-books, forms, official letters, admit cards, and result sheets, where the exact appearance matters.

Contrast this with a word-processing document, which is a *re-flowable* format: text re-flows to fit the screen, so the layout can shift when the font or device changes. A PDF, by design, does not re-flow; it is a picture of a finished page expressed as instructions.

Definition (PDF). *Portable Document Format — a fixed-layout file format that preserves the appearance of a document so it renders identically across devices and printers.*

33.6 Text and scanned PDFs

Not every PDF contains real text. It is important to separate two kinds.

A **text PDF** (sometimes called an editable-source PDF) is generated from a source document such as a Word file. Its pages contain genuine text and vector graphics. The text can be selected, copied, and searched, and the file is usually smaller.

A **scanned PDF** (image-only PDF) is produced by scanning printed pages or photographing them. Each page is stored as an *image*, much like a photograph. The text is readable to the human eye but not to the computer: it cannot be selected, copied, or searched. Scanned files are also usually larger.

Example 33.6.1. *An item says, “In a scanned PDF, you can search for a word using **Ctrl+F**.” This is wrong. A scanned PDF is only a picture of the page, so its text is not searchable until OCR is applied. In a text PDF, the same search works immediately.*

33.7 OCR: Optical Character Recognition

OCR stands for **Optical Character Recognition**. It is the technique that reads the *picture of the text* and converts it into **machine-readable text**. Once OCR has been run, the text in a scanned document can be selected, searched, copied, and edited just like the text in a text PDF.

OCR is used on scanned pages and on photographs of documents. It is the bridge that turns an image-only PDF into a searchable one. The recognition is not always perfect, so the result should be checked.

Definition (OCR (Optical Character Recognition)). *The process of converting an image of text into machine-readable, editable, and searchable text.*

33.8 Encryption and password protection

A PDF can be protected so that its contents are confidential. **Encryption** scrambles the file so that it cannot be read without the correct key or password. Two kinds of password are common. A **password to open** (user password) is required before the file will open at all. A **permissions password** (owner password) restricts what a reader may do — for example, printing, copying, or editing.

Encryption provides **confidentiality**: only the authorised reader can see the contents. It does not prove who created the file, and it is not the same as a digital signature.

33.9 Digital signature

A **digital signature** is attached to a file to prove **who signed it**. It also proves **integrity**: if the file is altered after it has been signed, the signature no longer matches and the change is detected. A digital signature uses a **digital certificate** issued by a trusted authority.

A digital signature provides **authenticity** (the signer is verified) and **integrity** (the file has not changed). It does *not* hide the contents. This is the distinction the exam most often tests: a digital signature is about *who* and *whether it changed*, while encryption is about *who can read it*.

Definition (Digital signature). *An electronic signature that verifies the identity of the signer and detects any change made to the file after signing, providing authenticity and integrity.*

Point	Encryption	Digital signature
Main purpose	Confidentiality	Authenticity and integrity
Result	Contents unreadable	Signer verified, tampering detected
Needs	Key / password	Digital certificate

Example 33.9.1. *An item states, “A digital signature encrypts the PDF so that only the recipient can read it.” The statement is wrong. The encryption that keeps others from reading the file is **encryption**. A digital signature does not hide the content; it confirms the signer and shows whether the file has been changed.*

33.10 Redaction and PDF/A

Redaction means **permanently removing** sensitive content before a file is shared. Names, account numbers, addresses, or photographs may be redacted. A proper redaction deletes the underlying text or image; it is not enough to draw a black rectangle over it, because hidden text can often still be selected and copied. A careful reader should always check that the removed text is truly gone.

PDF/A is a special ISO-standard version of PDF designed for **long-term archiving**. It embeds the fonts and colour information inside the file so that the document will render exactly the same years later, even on future software. The “A” in PDF/A stands for *archival*. It is used by libraries, courts, and government records, and is meant for preservation rather than everyday editing.

33.11 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. The **extension** is the file-name suffix; the **format** is the structure of the data. **.png** is an extension; **PNG** is the format (TRAP-34).
2. A **scanned PDF** is an image; its text cannot be selected or searched until **OCR** is applied. A text PDF is searchable at once.
3. **Encryption** gives confidentiality; a **digital signature** gives authenticity and integrity. They are not the same (TRM-11, TRAP-15).
4. **Redaction** removes sensitive content permanently; a black box drawn over text is not sufficient.
5. **PDF/A** is for long-term **archiving**; ordinary PDF is the everyday fixed-layout format.

Example 33.11.1. *An item asks which format keeps a fixed layout and looks the same on every device. The answer is PDF. If the stem instead asks which format stores spreadsheet data, the answer is XLSX; if it asks which is a plain-text data format, the answer is CSV; and if it asks which format is meant for long-term archiving, the answer is PDF/A.*

Exercises

1. Distinguish a file extension from a file format, with one example.
2. State the full form of PDF and explain why PDF is called a fixed-layout format.
3. Explain the difference between a text PDF and a scanned PDF, and where OCR fits in.
4. Distinguish encryption from a digital signature in terms of the security goal each provides.
5. What is redaction, and why is drawing a black box over text not a safe redaction? What is PDF/A used for?

Solutions

1. The extension is the dot-suffix in the file name, such as `.docx`; it is only a label. The format is the actual structure of the data inside the file, such as the DOCX word-processing format. The same extension can be renamed without changing the format.
2. PDF stands for Portable Document Format. It is called a fixed-layout format because every item is placed at a fixed position on the page, and the fonts and images are carried in the file, so the document looks the same on every device and printer.
3. A text PDF contains real, machine-readable text, so it can be selected, copied, and searched. A scanned PDF stores each page as an image, so its text cannot be selected or searched. OCR (Optical Character Recognition) converts the image of the text into machine-readable text, after which the scanned PDF becomes searchable.
4. Encryption provides confidentiality: it scrambles the file so only someone with the correct key or password can read it. A digital signature provides authenticity and integrity: it proves who signed the file and detects any change made after signing. It does not hide the contents.
5. Redaction is the permanent removal of sensitive content before a file is shared. Drawing a black box over text is not safe because the hidden text may still be selectable and copyable underneath. PDF/A is a standardised PDF version for long-term archiving that embeds fonts and colour information so the document renders the same in the future.